# Research Question Registry

Project Genesis Research Repository

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This registry documents the core scientific inquiries that Project Genesis intends to investigate. It links theoretical questions to the concrete laboratory instruments and experiments that provide the evidence.

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## RQ-001: Scarcity vs. Cooperation Dynamics

- \textbfInquiry: \textitCan stable cooperative attachment and sharing mechanisms emerge in environments constrained by high selection pressure?
- \textbfHypothesis: If resource scarcity exceeds critical thresholds (scarcity > 4x), territorial disputes will rise, suppressing mutual attachment and reproduction rates, leading to early lineage extinction cascades.
- \textbfSupported Instruments:
- \texttttOBS-001 (Population Dynamics)
- \texttttOBS-003 (Territorial Friction Map)
- \texttttOBS-004 (Social Kinship Grid)
- \textbfExperiments:
- \texttttEXP-20260714-H4K9 (The Cooperation Paradox Under Scarcity)

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## RQ-002: Genetic Entropy & Selection Pressures

- \textbfInquiry: \textitDoes high genetic mutation speed accelerate adaptability or trigger genetic collapse under volatile climate epochs?
- \textbfHypothesis: Higher mutation rates (std > 0.08) accelerate early biome adaptation, but lead to rapid genetic drift and survival collapse when the climate enters a prolonged stable epoch.
- \textbfSupported Instruments:
- \texttttOBS-001 (Population Dynamics)
- \texttttOBS-005 (Genetic Entropy Browser)
- \textbfExperiments:
- TBD (Pending Phase 4 runs)

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## RQ-003: Settlement Mergers & Spitting (Colony Trees)

- \textbfInquiry: \textitHow do resource carrying capacities influence colony fission and fusion dynamics?
- \textbfHypothesis: Colonies split (fission) when localized water sources deplete below 30% capacity, but merge (fusion) when single, high-capacity reserves are shared.
- \textbfSupported Instruments:
- \texttttOBS-001 (Population Dynamics)
- \texttttOBS-004 (Social Kinship Grid)
- \textbfExperiments:
- TBD